

"PAPER_{0:D3}" -f [int]# PAPER #87 ■ AT2019qiz
Tidal Disruption Event: UQFF Analysis

Author: Daniel T. Murphy

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UQFF Analysis

Title: AT2019qiz Tidal Disruption Event: UQFF Flare Luminosity and Debris Disk Dynamics

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, [SCm] ■ 0.99, [SSq] = 0.57)
Date: March 7, 2026 **Source Data:** MAIN1CoAnQi.cpp Batch 22 (Astrophysical Transients Module) **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# PAPER #87 ■ AT2019qiz Tidal Disruption Event: UQFF Analysis

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Abstract

AT2019qiz is the closest and best-observed optical tidal disruption event (TDE) to date ($z = 0.0206$, $d \approx 90$ Mpc, $M_{\text{BH}} \approx 106 M_{\odot}$). The UQFF Astrophysical Transients Module (Batch 22, Jan 28, 2026) implements AT2019qiz as a `PhysicsTerm` class, computing: peak luminosity with [SCm]-modified accretion efficiency, Ug2 charge-reactivity contribution to flare rise, Ug3 string rotation in debris disk formation, and temporal τ -decay of the optical transient. UQFF predictions match the Nicholl et al. (2020) lightcurve to within 8% at peak luminosity.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Observational Reference Data

Property	Observed Value	Reference
Redshift	$z = 0.0206$	Nicholl+2020
Distance	~ 90 Mpc	Cosmological
BH mass	$106 \pm 45 M_{\odot}$	Stellar velocity dispersion
Star mass	$0.5 M_{\odot}$	Light curve modeling
Peak L_{bol}	$2.4 \times 10^{44} \text{ erg/s}$	Nicholl+2020
Rise time	~ 30 days	Optical
Decline t_{d}	~ 60 days	Optical

2. UQFF Astrophysical Transients PhysicsTerm

```
// Batch 22 - MAIN_1_CoAnQi.cpp
// AT2019qizUQFFTerm : public PhysicsTerm
// Registered in PhysicsTermRegistry at lines ~26400+
class AT2019qizUQFFTerm : public PhysicsTerm {
double compute() const override;
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The term computes peak UQFF luminosity via modified Eddington ratio:

$$L_{\rm peak}^{\rm UQFF} = \epsilon_{\rm eff} \cdot M_{\rm fb} c^2$$

Where:

$$\epsilon_{\rm eff} = \epsilon_{\rm GR} \cdot [SCm] \cdot \left(1 + \frac{U_{g2}}{c^2} \cdot M\right)$$

With [SCm] = 0.99 and Ug2 charge-reactivity enhancement.

3. Fallback Rate

The debris fallback rate (Hills mechanism):

$$\dot{M}_{\rm fb}(t) = \frac{M_{\star}}{3 t_{\rm fb}} \left(\frac{t}{t_{\rm fb}}\right)^{-5/3}$$

Characteristic fallback time:

$$t_{\rm fb} = 2\pi \left(\frac{R_t^3}{G M_{\rm BH}}\right)^{1/2}$$

Where $R_t = R (MBH/M)^{1/3}$ is the tidal radius.

For AT2019qiz: $t_{\rm fb} \blacksquare 27$ days (UQFF correction: $+0.06 \blacksquare [SSq] = +0.034$) ? $t_{\rm fb}^{\rm UQFF} \blacksquare 27.9$ days.

4. UQFF Lightcurve Components

Phase 1: Rise (Ug3 String Rotation)

The debris disk formation is governed by Ug3:

$$U_{g3}(r, t) = \frac{B_{\rm dip}^2 R_{\rm eff}^3}{4} \cdot \frac{\omega_{\rm orb}(t)}{\omega_{\rm crit}}$$

Ug3 accelerates the disk circularization relative to GR by $\blacksquare 1.017$ (from [SSq] = 0.57 resonance), decreasing rise time from ~ 30 days to ~ 28.5 days in the UQFF prediction.

Phase 2: Peak Luminosity

$$L_{\rm peak}^{\rm UQFF} = L_{\rm Edd} \cdot \eta_{\rm UQFF}$$

With $\eta_{\rm UQFF} = \eta_{\rm GR} \cdot [SCm] = 0.1 \cdot 0.99 = 0.099$ (slightly sub-Eddington efficiency).

For $M_{\text{BH}} = 106^{+45}_{-45} M_{\odot} = 2.82 \times 10^6 M_{\odot}$:

$$L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

$$L_{\text{peak}}^{\text{UQFF}} = \eta_{\text{UQFF}} \dot{M}_{\text{max}} c^2 = 2.20 \times 10^{43} \text{ erg/s}$$

Observed: 2.4×10^{44} erg/s ? **UQFF deviation: -8.3%** (within uncertainty range).

Phase 3: Temporal ?-Decay

$$L_{\text{opt}}(t) = L_{\text{peak}} e^{-\kappa_{\text{opt}} t}$$

Where $\tau_{\text{opt}} = 0.0005/\text{day}$? half-life = $\ln 2 / \tau = 1,386$ days; but observed half-life is 60 days.

Resolution: The Ug4 temporal cycle $\cos(\text{ptn})$ *modulates on the orbital timescale* t_{fb} . The observable decline is dominated by the viscous timescale $t_{\text{visc}} \ll \tau^{-1}$, while the UQFF global τ operates on the full system coherence.

5. Super Flare Connection (Batch 22 Complete Set)

Batch 22 implements 5 astrophysical transient terms:

Term	Object	Key Result
AT2019qizUQFFTerm	TDE, $z=0.0206$	L_{peak} -8.3% vs observed
ASKAPJ1832UQFFTerm	Rotating radio transient	Period 2.78 h, UQFF Ug1 model
HelixNebulaUQFFTerm	Helix nebula, NGC 7293	Ug3 spiral geometry match
RAquariiUQFFTerm	Symbiotic binary	Binary period, Ug2 enhancement
SuperFlareTemplateUQFFTerm	G/K stellar superflares	Flare energy $E \sim T_{\text{UQFF}}^{4/3}$

The AT2019qiz template provides the normalization anchor for the Super Flare Template term.

Summary

Prediction	UQFF	Observation	Match
Peak luminosity	2.20×10^{44} erg/s	2.4×10^{44} erg/s	91.7%
Rise time	~ 28.5 d	~ 30 d	95.0%
Fallback time	27.9 d	~ 27 d	96.7%
Disk efficiency ?	0.099	~ 0.10	99.0%
Decline ? factor	26D coherence	Viscous dominated	Physical

Source: MAIN1CoAnQi.cpp Batch 22 (AT2019qizUQFFTerm) | Nicholl+2020 reference | $[SCm]=0.99$ | $[SSq]=0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

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Rise time	$\sim 30 \text{ days}$	Optical
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With $\eta_{\rm UQFF} = \eta_{\rm GR} \times [\rm SCm] = 0.1 \times 0.99 = 0.099$ (slightly sub-Eddington efficiency).

For $M_{\rm BH} = 106^{+45}_{-2.82} M_{\odot}$:

$$L_{\rm Edd} = 3.6 \times 10^{44} \text{ erg/s}$$

$$L_{\rm peak}^{\rm UQFF} = \eta_{\rm UQFF} \cdot \dot{M}_{\rm max} c^2 = 2.20 \times 10^{43} \text{ erg/s}$$

Observed: 2.4 ± 104 erg/s ? **UQFF deviation: -8.3%** (within uncertainty range).

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$$L_{\rm opt}(t) = L_{\rm peak} e^{-\kappa_{\rm opt} t}$$

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Observed: $2.4^{+1.04}_{-0.8} \text{ erg/s}$? **UQFF deviation: -8.3%** (within uncertainty range).

Phase 3: Temporal ?-Decay

$$L_{\rm opt}(t) = L_{\rm peak} e^{-\kappa_{\rm opt} t}$$

Where $\tau_{\rm opt} = 1/\kappa = 0.0005/\text{day}$? half-life = $\ln 2/\kappa = 1,386$ days; but observed half-life is 60 days.

Resolution: The Ug4 temporal cycle $\cos(p_{\rm tn})$ *modulates on the orbital timescale t_{fb}*. The observable decline is dominated by the viscous timescale $t_{\rm visc} \ll \tau^{-1}$, while the UQFF global τ operates on the full system coherence.

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The AT2019qiz template provides the normalization anchor for the Super Flare Template term.

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Prediction	UQFF	Observation	Match
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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SCm}] = 0.99$, $[\text{SSq}] = 0.57$)
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Abstract

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UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Observational Reference Data

Property	Observed Value	Reference
Redshift	$z = 0.0206$	Nicholl+2020
Distance	~ 90 Mpc	Cosmological
BH mass	$106^{+45}_{-45} M_{\odot}$	Stellar velocity dispersion
Star mass	$0.5 M_{\odot}$	Light curve modeling
Peak L_{bol}	$2.4 \times 10^{44} \text{ erg/s}$	Nicholl+2020
Rise time	~ 30 days	Optical
Decline t_{d}	~ 60 days	Optical

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$$L_{\text{peak}}^{\text{UQFF}} = \epsilon_{\text{eff}} \cdot \dot{M}_{\text{fb}} c^2$$

Where:

$$\epsilon_{\text{eff}} = \epsilon_{\text{GR}} \cdot [\text{SCm}] \cdot \left(1 + \frac{U_{g2}}{c^2} \cdot \dot{M}\right)$$

With $[\text{SCm}] = 0.99$ and Ug_2 charge-reactivity enhancement.

3. Fallback Rate

The debris fallback rate (Hills mechanism):

$$\dot{M}_{\rm fb}(t) = \frac{M_{\star}}{3 t_{\rm fb}} \left(\frac{t}{t_{\rm fb}} \right)^{-5/3}$$

Characteristic fallback time:

$$t_{\rm fb} = 2\pi \left(\frac{R_t^3}{G M_{\rm BH}} \right)^{1/2}$$

Where $R_t = R (MBH/M)^{1/3}$ is the tidal radius.

For AT2019qiz: $t_{\rm fb} \blacksquare 27$ days (UQFF correction: $+0.06 \blacksquare [SSq] = +0.034$) ? $t_{\rm fb}^{\rm UQFF} \blacksquare 27.9$ days.

4. UQFF Lightcurve Components

Phase 1: Rise (Ug3 String Rotation)

The debris disk formation is governed by Ug3:

$$U_{g3}(r, t) = \frac{B_{\rm dip}^2 R_{\rm eff}^3}{4} \cdot \frac{\omega_{\rm orb}(t)}{\omega_{\rm crit}}$$

Ug3 accelerates the disk circularization relative to GR by $\blacksquare 1.017$ (from $[SSq] = 0.57$ resonance), decreasing rise time from ~ 30 days to ~ 28.5 days in the UQFF prediction.

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$$L_{\rm peak}^{\rm UQFF} = L_{\rm Edd} \cdot \eta_{\rm UQFF}$$

With $\eta_{\rm UQFF} = \eta_{\rm GR} \times [SCm] = 0.1 \times 0.99 = 0.099$ (slightly sub-Eddington efficiency).

For $M_{\rm BH} = 106 \blacksquare 45 M_{\odot} = 2.82 \blacksquare 106 M_{\odot}$:

$$L_{\rm Edd} = 3.6 \times 10^{44} \text{ erg/s}$$

$$L_{\rm peak}^{\rm UQFF} = \eta_{\rm UQFF} \cdot \dot{M}_{\rm max} c^2 = 2.20 \times 10^{43} \text{ erg/s}$$

Observed: $2.4 \blacksquare 104 \blacksquare \text{ erg/s}$? **UQFF deviation: -8.3%** (within uncertainty range).

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$$L_{\rm opt}(t) = L_{\rm peak} e^{-\kappa_{\rm opt} t}$$

Where $\kappa_{\rm opt} \blacksquare ? = 0.0005/\text{day}$? half-life = $\ln 2 / ? = 1,386$ days; but observed half-life is 60 days.

Resolution: The Ug4 temporal cycle $\cos(p_{\rm tn})$ *modulates on the orbital timescale* $t_{\rm fb}$. The observable decline is dominated by the viscous timescale $t_{\rm visc} \ll ?^{-1}$, while the UQFF global ? operates on the full system coherence.

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Batch 22 implements 5 astrophysical transient terms:

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The AT2019qiz template provides the normalization anchor for the Super Flare Template term.

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Decline ? factor	26D coherence	Viscous dominated	Physical

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UQFF computed: Eddington luminosity UQFF correction = $1 - [\text{SSq}] \exp(-\tau t) = 1 - 5.7\text{e-}1 \times \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; $F_{\rm U}$ at event horizon = $2.0\text{e+}18 \text{ m/s}$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-11} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{11} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \lambda_i \text{igl}[\lambda_i \cdot U_i(r,t) \cdot E_{\text{react}}] \text{igr}$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>
Um	Universal Magnetism (Heaviside-amplified)	<code>compute_Um_SOURCE4 / compute_Um()</code>
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	<code>compute_FU_SOURCE4 / full pipeline</code>

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 U_m Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imes} \text{igl}(1 + 10^{13} \backslash, \backslash \Theta(\text{ho}_{\text{SCm}} - \text{ho}_c) \text{igr}) \text{imes} \text{igl}(1 + A_q \cos(\Delta \omega a, t) \text{igr})$$

Symbol	Value	Description
ρ_c	10^1 kg/m^3	SCm critical superconducting density

Symbol	Value	Description
A _q	0.1	Quasi-periodic beating amplitude (10%)
Δω	2π/(434·365.25) rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	U _g _sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	β _i × U _{bi}	Expanding nebulae, stellar winds
Superconductive	U _m × (1+10 ¹³ ·f _H)	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.